

Week11-12_MultivariateAnalysis

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4.1 Multivariate Resemblance

4.1.1 Binary Similarity Metrics

```
species <- c(1:24)
erie <- c(rep(1, 19), rep(0,5))
ontario <- c(rep(1,18), 0, 1, 0,0,0,0)
lksp <- data.frame(species = species,erie = erie, ontario = ontario)

#fixes ordering problem
lksp$erie <- factor(lksp$erie, levels = c(1, 0))
lksp$ontario <- factor(lksp$ontario, levels = c(1, 0))

tlake <- table(lksp[, c("erie", "ontario")])

a = tlake[1, 1]
b = tlake[1, 2]
c = tlake[2, 1]
d = tlake[2, 2]

S_j = a/(a + b + c)
S_j
```

```
## [1] 0.9
```

```
S_s = 2 * a/(2 * a + b + c)
S_s
```

```
## [1] 0.9473684
```

4.1.2.1 Euclidean Distance

Exercise 1 Finish the calculation of the Euclidean distance for this data.

```

#create the dataset
species <- paste("Species", 1:10)
sample_j <- c(19, 35, 0, 35, 10, 0, 0, 0, 30, 2)
sample_k <- c(35, 10, 0, 5, 50, 0, 3, 0, 10, 0)

eu <- data.frame(
  species = species,
  j = sample_j,
  k = sample_k,
  sq_diff = (sample_j - sample_k)^2
)

#calculate Euclidean distance
dist(rbind(eu$j, eu$k), method = "euclidean")

##          1
## 2 61.59545

```

4.2 Cluster Analysis

4.2.4 Example: Cluster analysis of isotope data

```

#install.packages("openxlsx")
library(openxlsx)

## Warning: package 'openxlsx' was built under R version 4.5.3

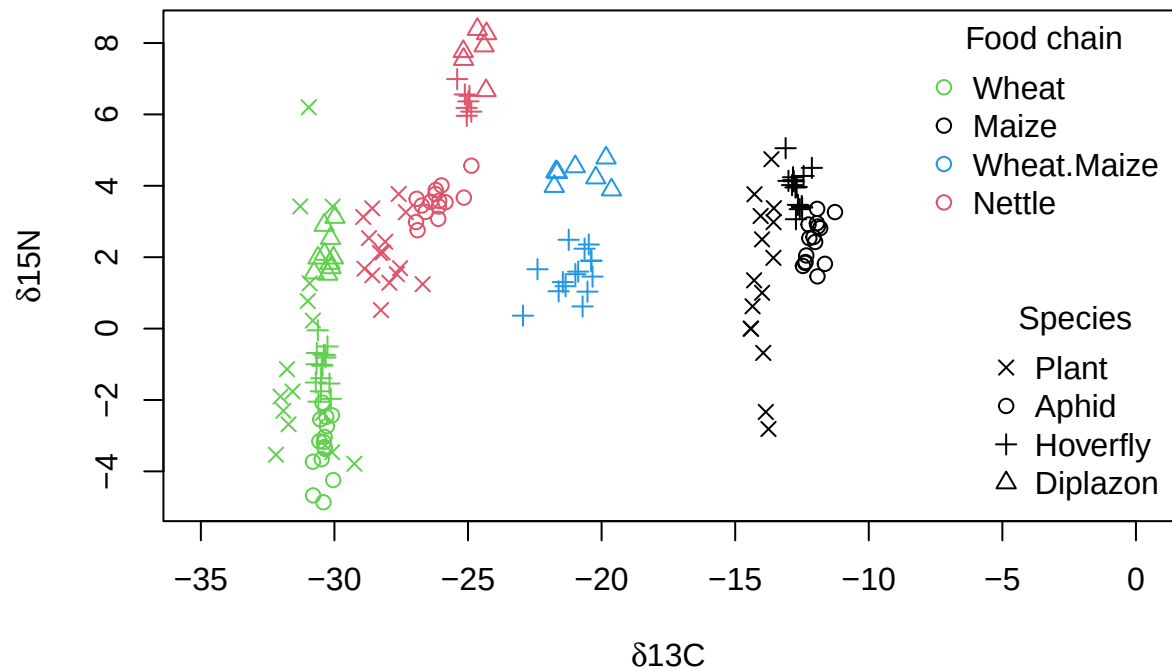
urlj = "https://doi.org/10.1371/journal.pone.0093281.s001"
download.file(urlj, "p.xlsx", mode = "wb")
iso = read.xlsx("p.xlsx")

plot(iso$N ~ iso$C, col = as.numeric(as.factor(iso$Food.Chain)),
     xlim = c(-35, 0), pch = as.numeric(as.factor(iso$Species)),
     xlab = expression(paste(delta, "13C")), ylab = expression(paste(delta,
       "15N")))

legend("topright", legend = unique(as.factor(iso$Food.Chain)),
     pch = 1, col = as.numeric(unique(as.factor(iso$Food.Chain))),
     bty = "n", title = "Food chain")

legend("bottomright", legend = as.character(unique(as.factor(iso$Species))),
     pch = as.numeric(unique(as.factor(iso$Species))), bty = "n",
     title = "Species")

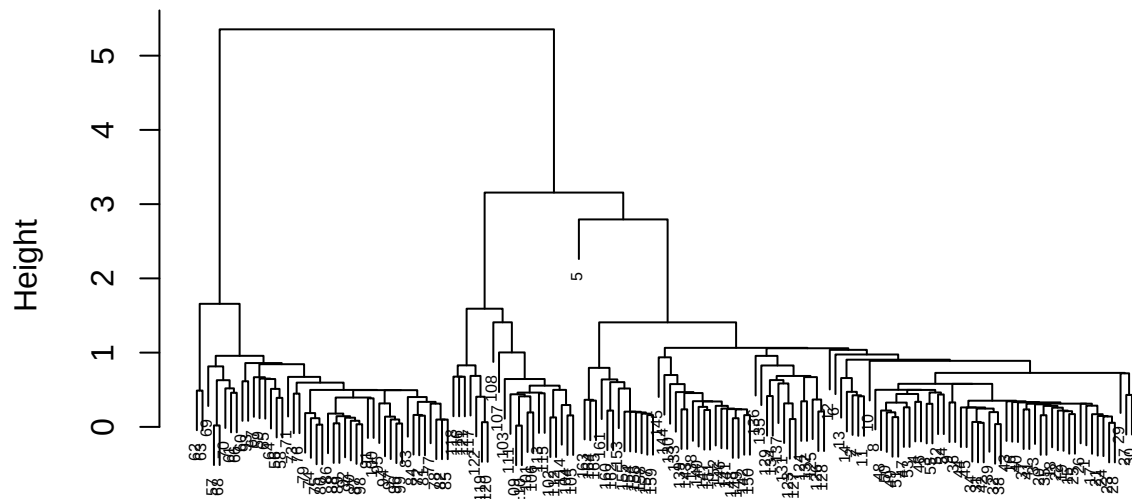
```



```
str(iso)
```

```
## 'data.frame': 165 obs. of 7 variables:
## $ Replicate : num 1 2 3 4 5 6 7 8 9 10 ...
## $ Food.Chain : chr "Wheat" "Wheat" "Wheat" "Wheat" ...
## $ Species : chr "Plant" "Plant" "Plant" "Plant" ...
## $ Tissue : chr "Leaf" "Leaf" "Leaf" "Leaf" ...
## $ Lipid.Extracted: chr "No" "No" "No" "No" ...
## $ C : num -30.1 -31.7 -30.1 -30.9 -31 ...
## $ N : num -3.47 -2.68 3.42 1.27 6.2 ...
```

```
diso <- dist((iso[, c("C", "N")]), method = "euclidean")
p = hclust(diso, method = "single", )
plot(p, cex = 0.5, main = "", xlab = "")
```



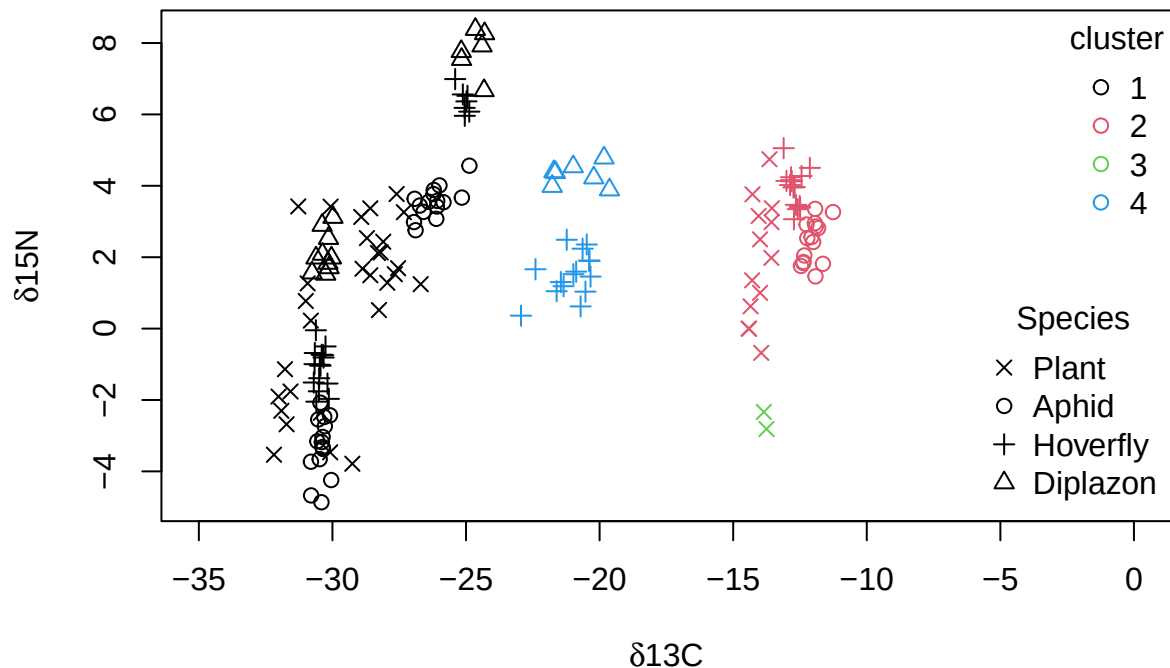
`hclust (*, "single")`

```
niso = niso[complete.cases(niso), ]
niso = niso[-5, ]
diso <- dist((niso[, c("C", "N")]), method = "euclidean")
p = hclust(diso, method = "single")
niso$clust <- cutree(p, k = 4)

# plotting the data with 4 groups identified by the
# single-linkage cluster analysis superimposed
plot(niso$N ~ niso$C, col = as.numeric(as.factor(niso$clust)),
     xlim = c(-35, 0), pch = as.numeric(as.factor(niso$Species)),
     xlab = expression(paste(delta, "13C")), ylab = expression(paste(delta,
"15N")))

legend("topright", legend = unique(as.factor(niso$clust)), pch = 1,
     col = as.numeric(unique(as.factor(niso$clust))), bty = "n",
     title = "cluster")

legend("bottomright", legend = as.character(unique(as.factor(niso$Species))),
     pch = as.numeric(unique(as.factor(niso$Species))), bty = "n",
     title = "Species")
```



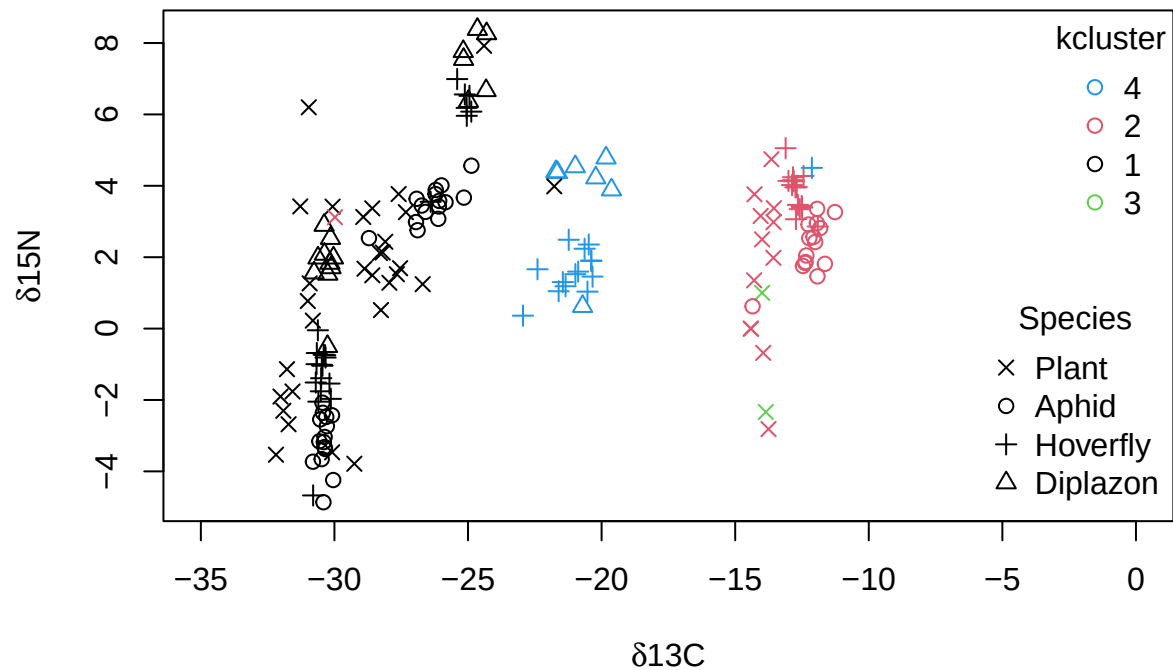
Exercise 2 Try a non-hierarchical cluster analysis on the same data to see if it works better. Use the `kmeans()` function, which requires that we select the required number of clusters (4) ahead of time. Once you have your clustering, plot your results.

```
kclust <- kmeans(niso[, c("C", "N")], 4)
niso$kclust = kclust$cluster

# superimpose the saved clusters on our plotted isotope
# data
plot(niso$N ~ niso$C, col = as.numeric(as.factor(niso$kclust)),
     xlim = c(-35, 0), pch = as.numeric(as.factor(niso$Species)),
     xlab = expression(paste(delta, "13C")), ylab = expression(paste(delta,
"15N")))

legend("topright", legend = unique(as.factor(niso$kclust)), pch = 1,
      col = as.numeric(unique(as.factor(niso$kclust))), bty = "n",
      title = "kcluster")

legend("bottomright", legend = as.character(unique(as.factor(niso$Species))),
      pch = as.numeric(unique(as.factor(niso$Species))), bty = "n",
      title = "Species")
```



4.3.4 Example: nMDS and PCoA

```
library(vegan)
```

```
## Warning: package 'vegan' was built under R version 4.5.2
```

```
## Loading required package: permute
```

```
## Warning: package 'permute' was built under R version 4.5.2
```

```
data(varespec)
nMDS <- metaMDS(varespec, trymax = 100, distance = "bray", k = 2,
  trace = FALSE)
svarespec = wisconsin(varespec)
disimvar = vegdist(svarespec, method = "bray")

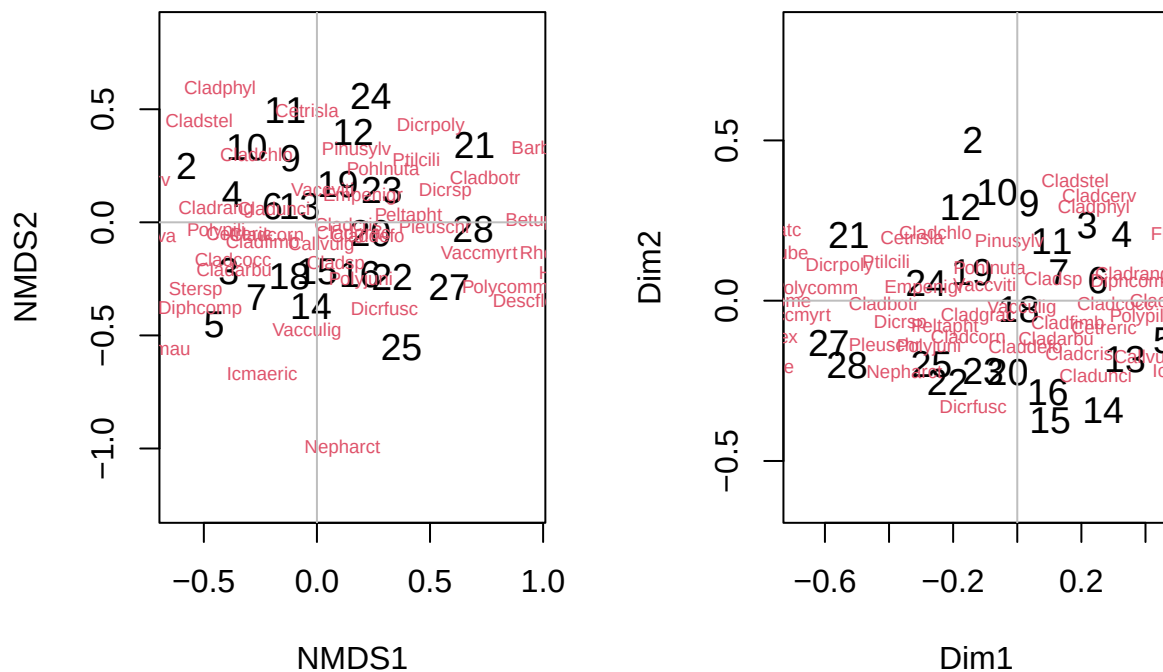
PCoA <- cmdscale(disimvar, k = 2, eig = T, add = T)
str(PCoA)
```

```
## List of 5
## $ points: num [1:24, 1:2] 0.118 0.103 -0.182 -0.486 -0.106 ...
## .. attr(*, "dimnames")=List of 2
## .. ..$ : chr [1:24] "18" "15" "24" "27" ...
## .. ..$ : NULL
## $ eig : num [1:24] 1.208 0.832 0.743 0.491 0.461 ...
```

```
## $ x      : NULL
## $ ac      : num 0.178
## $ GOF     : num [1:2] 0.298 0.298
```

```
par(mfrow = c(1, 2))
ordipointlabel(nMDS, pch = c(NA, NA), cex = c(1.2, 0.6), xlim = c(-0.6,
1.2))
abline(h = 0, col = "grey")
abline(v = 0, col = "grey")

PCoA$species <- wascores(PCoA$points, varespec, expand = TRUE)
ordipointlabel(PCoA, pch = c(NA, NA), cex = c(1.2, 0.6), xlab = "PCoA1",
ylab = "PCoA2", xlim = c(-0.6, 1), ylim = c(-0.5, 0.6))
abline(h = 0, col = "grey")
abline(v = 0, col = "grey")
```

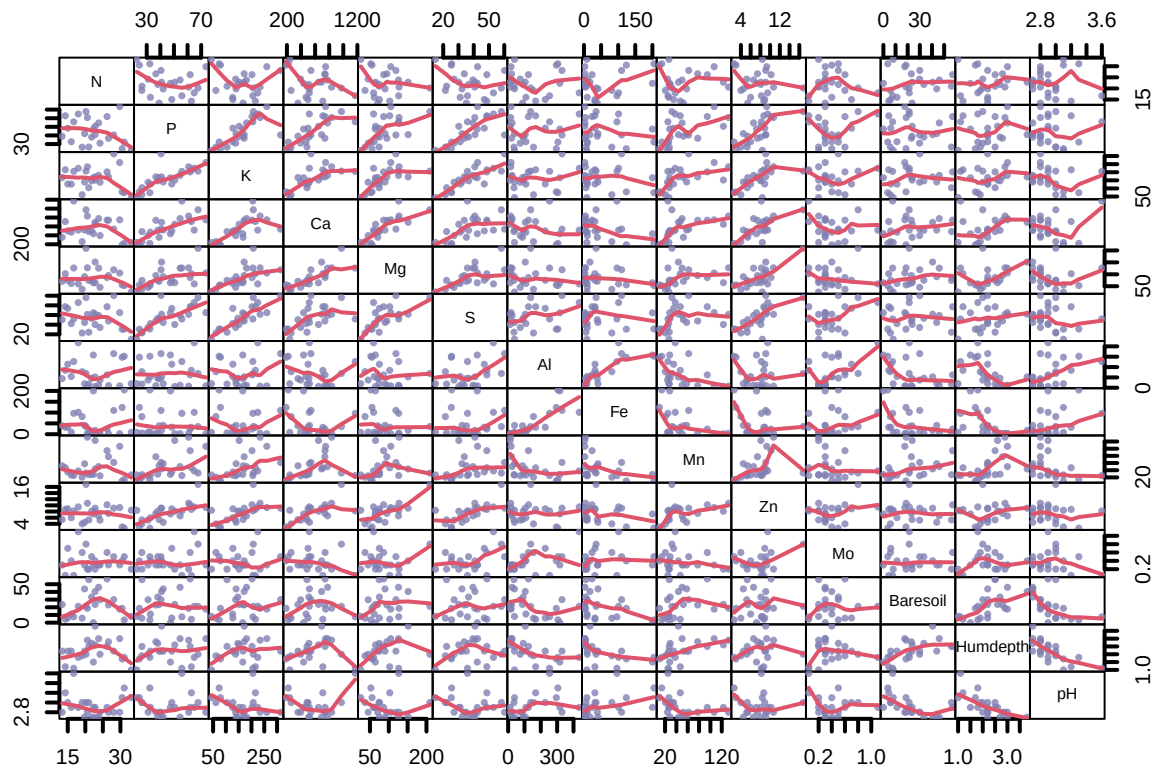


Exercise 3 Provide an interpretation of the above plots > **Answer 3:** Sites (numbers) that are closer to one another are more similar. The species (red text) are distributed based on their influence on these patterns. This is most easily visible by sites that are more isolated from others and that have a species that is only aligned near them. For example, site 2 may be more different due to *Cladocerv* spp.

4.4.2 Example: Constrained ordination

```
library(vegan)
data("varespec")
data("varechem")
```

```
plot(varechem, gap = 0, panel = panel.smooth, cex.lab = 1.2,
     lwd = 2, pch = 16, cex = 0.75, col = rgb(0.5, 0.5, 0.7, 0.8))
```



```
stvarespec = as.data.frame(wisconsin(varespec))
scvarechem = as.data.frame(scale(varechem))
constr <- vegan::rda(stvarespec ~ ., data = scvarechem)
constr
```

```
##
## Call: rda(formula = stvarespec ~ N + P + K + Ca + Mg + S + Al + Fe + Mn +
## Zn + Mo + Baresoil + Humdepth + pH, data = scvarechem)
##
##              Inertia Proportion Rank
## Total          0.05084    1.00000
## Constrained    0.03612    0.71057   14
## Unconstrained  0.01471    0.28943    9
##
## Inertia is variance
```

```
##
## Eigenvalues for constrained axes:
##      RDA1      RDA2      RDA3      RDA4      RDA5      RDA6      RDA7      RDA8
## 0.008120 0.005642 0.004808 0.003525 0.002956 0.002592 0.001953 0.001665
##      RDA9      RDA10     RDA11     RDA12     RDA13     RDA14
## 0.001265 0.001129 0.000964 0.000627 0.000507 0.000373
##
## Eigenvalues for unconstrained axes:
##      PC1      PC2      PC3      PC4      PC5      PC6      PC7      PC8
## 0.004051 0.002184 0.002003 0.001853 0.001247 0.001101 0.001022 0.000722
##      PC9
## 0.000532
```

```
ord3 = vegan::rda(stvarespec ~ N + K + Al, data = scvarechem)
ord3
```

```
##
## Call: rda(formula = stvarespec ~ N + K + Al, data = scvarechem)
##
##              Inertia Proportion Rank
## Total          0.05084    1.00000
## Constrained    0.01171    0.23041    3
## Unconstrained 0.03913    0.76959   20
##
## Inertia is variance
##
## Eigenvalues for constrained axes:
##      RDA1      RDA2      RDA3
## 0.006722 0.003232 0.001761
##
## Eigenvalues for unconstrained axes:
##      PC1      PC2      PC3      PC4      PC5      PC6      PC7      PC8
## 0.007246 0.004097 0.003737 0.003515 0.003305 0.002474 0.002369 0.001785
## (Showing 8 of 20 unconstrained eigenvalues)
```

4.4.3 Significance tests for constrained ordination

```
anova(ord3)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = stvarespec ~ N + K + Al, data = scvarechem)
##      Df Variance      F Pr(>F)
## Model    3 0.011714 1.996 0.001 ***
## Residual 20 0.039125
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(ord3, by = "term", permutations = 199)
```

```
## Permutation test for rda under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 199
##
## Model: rda(formula = stvarespec ~ N + K + Al, data = scvarechem)
##           Df Variance      F Pr(>F)
## N           1 0.002922 1.4937 0.070 .
## K           1 0.003120 1.5948 0.090 .
## Al          1 0.005672 2.8994 0.005 **
## Residual    20 0.039125
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
mfull <- vegan::rda(stvarespec ~ ., data = scvarechem)
m0 <- vegan::rda(stvarespec ~ 1, data = scvarechem)

optm <- ordiR2step(m0, scope = formula(mfull), trace = FALSE)
optm$anova
```

```
##           R2.adj Df      AIC      F Pr(>F)
## + Fe          0.063299 1 -71.156 2.5543 0.002 **
## + P           0.103626 1 -71.329 1.9898 0.008 **
## + Mn          0.136721 1 -71.402 1.8051 0.022 *
## <All variables> 0.260356
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Exercise 4 Examine the significance test. How does this automatically selected model differ from the full model and your manually selected version? > **Answer 4:** From the full model's 14 elements, we manually selected N, K, and Al. These elements do not appear in the automatically selected model, which identified Fe, P, and Mn as the most significant predictors.